EVPNユースケース

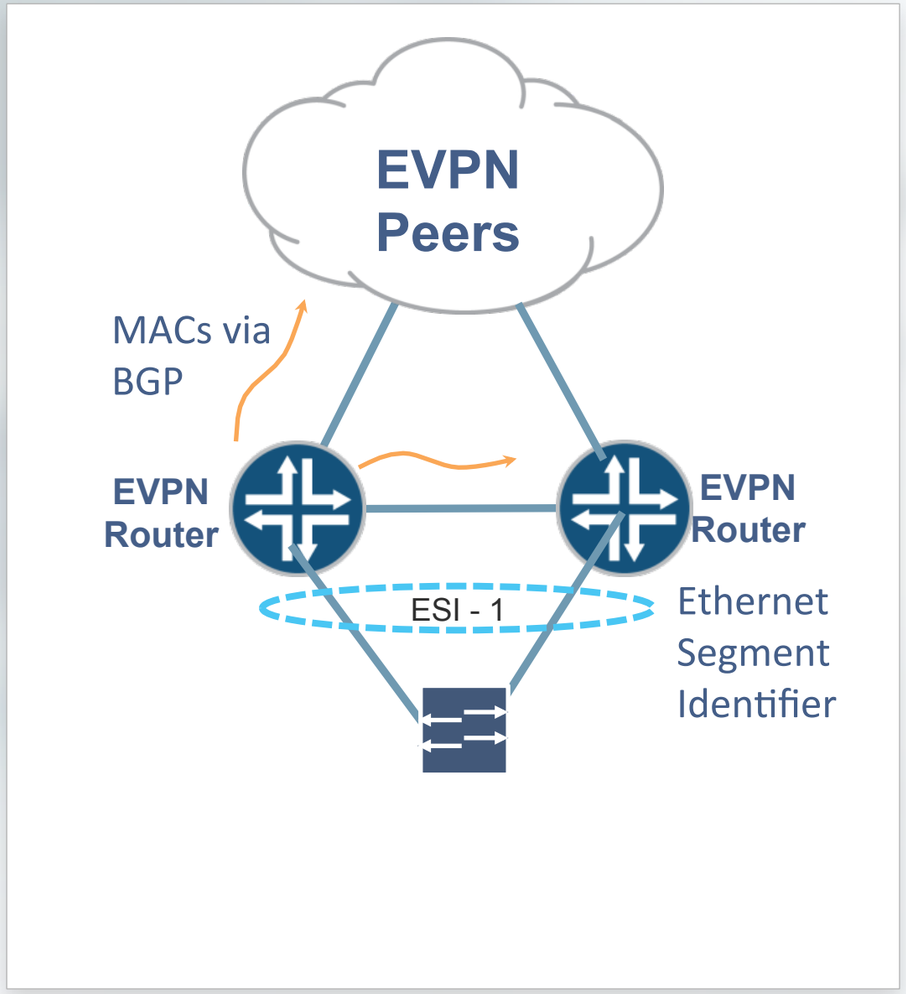
データセンターでの活用

ジュニパーネットワークス株式会社
長滝　信彦

EVPNの適用領域

本日の話題

Data Center	All Active load balancing. Seamless L2/L3 Interconnect
Business Edge	Multi-homed, feature rich E-LINE, E-LAN E-TREE
Metro	Multi-homing and L3 Integration
Peering	Policy driven peering relationships
OTT	L2/L3 Services over simple IP connectivity



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EVPNの利点

Custom Services

IPVPN like policy support for service customization

Link Efficiency

All Active forwarding with L2 Loop Prevention

Convergence

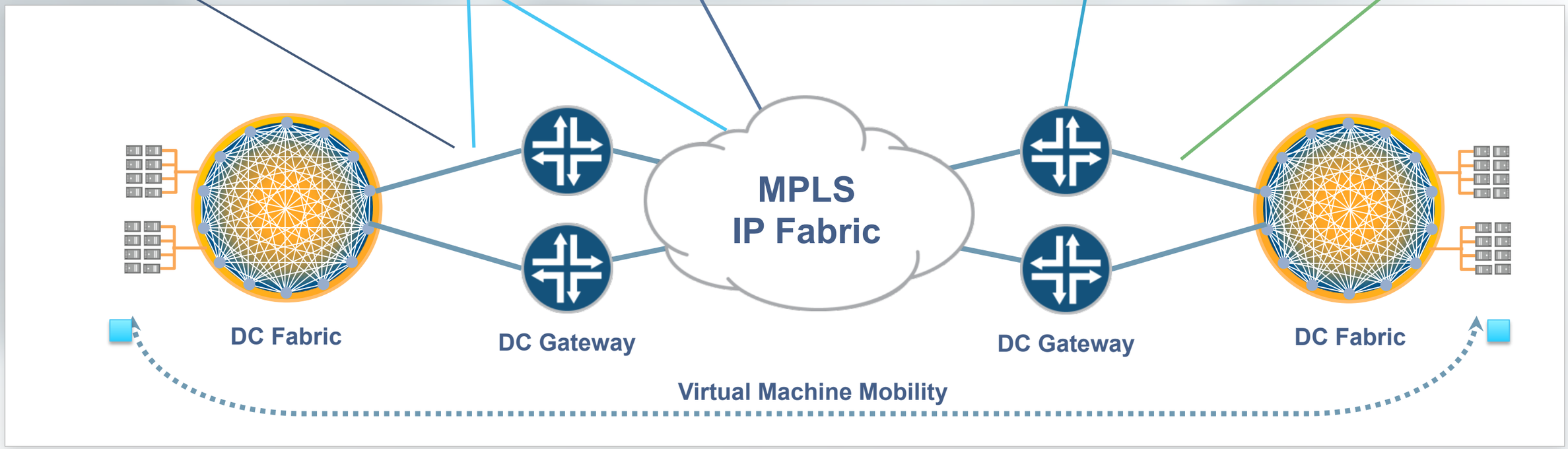
Built in HA - fast convergence, fast reroute capabilities

L3 and L2

L2 & L3 Layers Tie-In Built-in the protocol

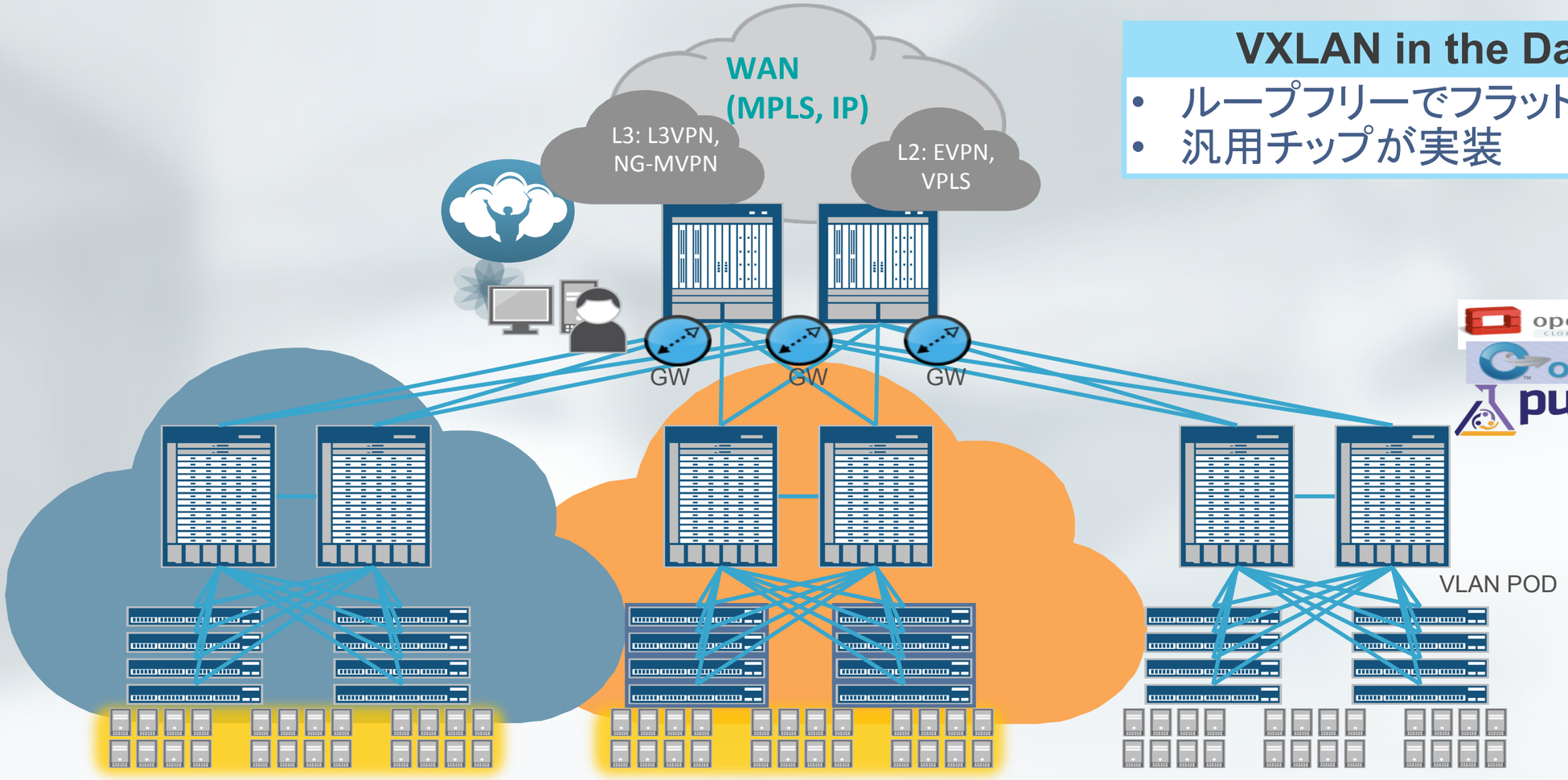
DC Optimized

Ingress and Egress VM Mobility Optimizations



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データセンター内の仮想化技術



VXLAN in the Data Center

- ループフリーでフラットなIPファブリック
- 汎用チップが実装

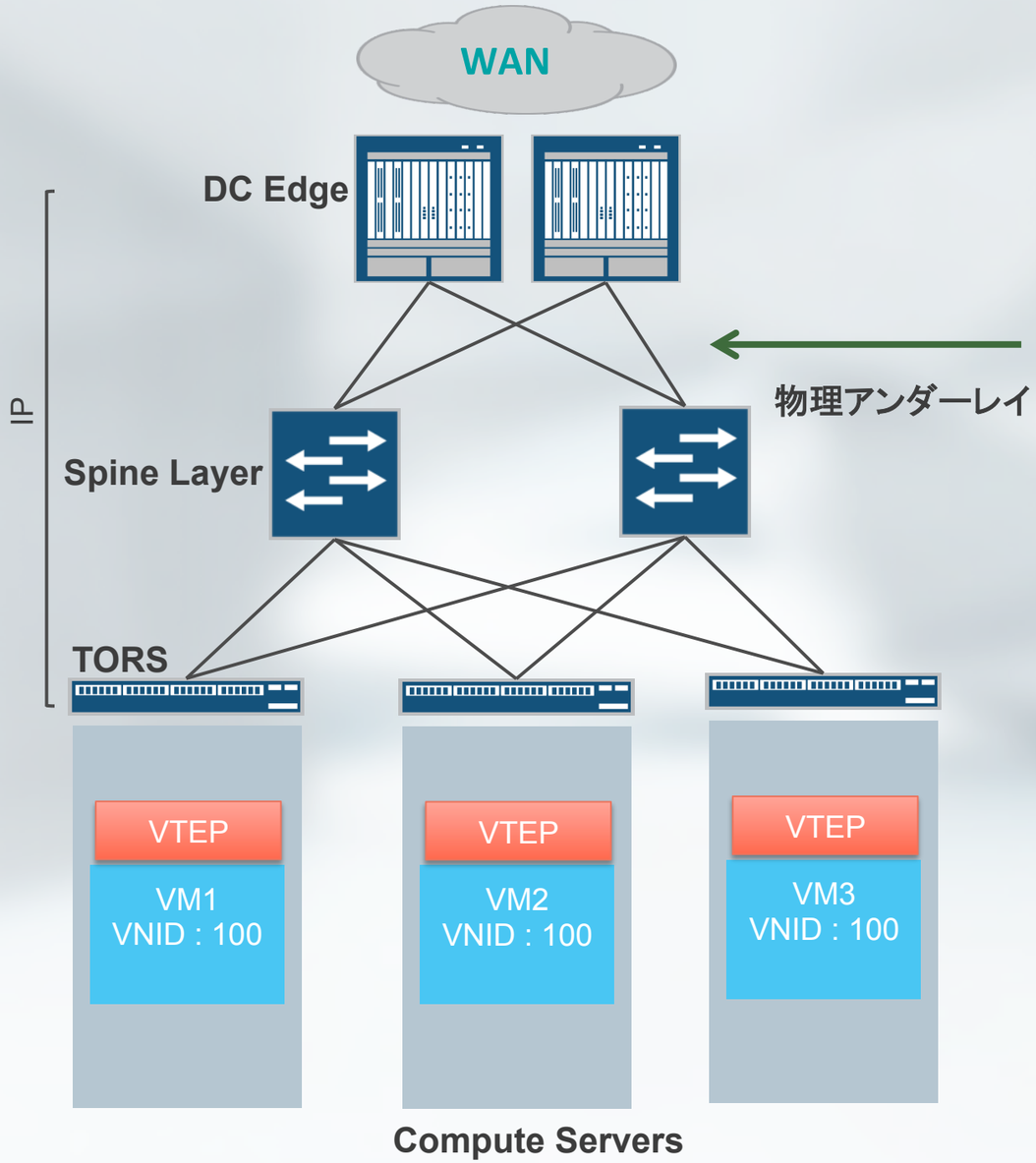
VXLAN Based POD

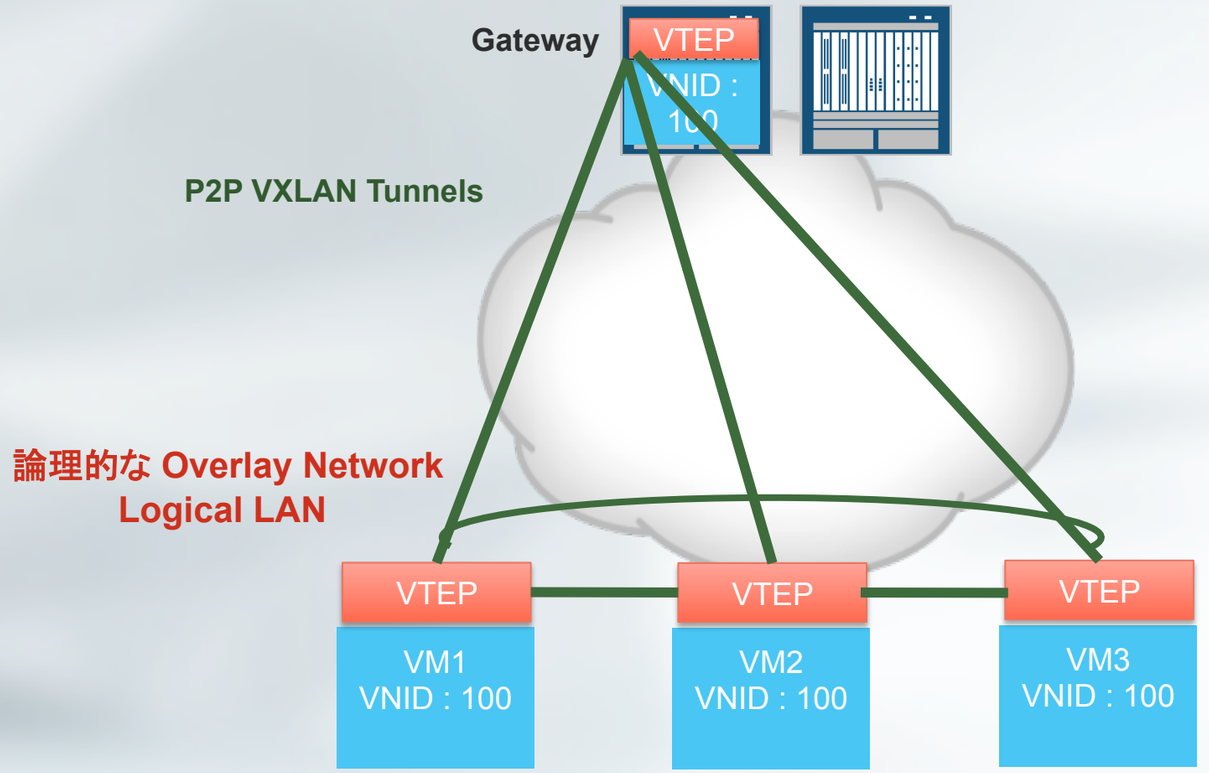
VXLAN based POD

Legacy, VLAN based POD

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VXLAN: Virtual Overlay and Physical Underlay



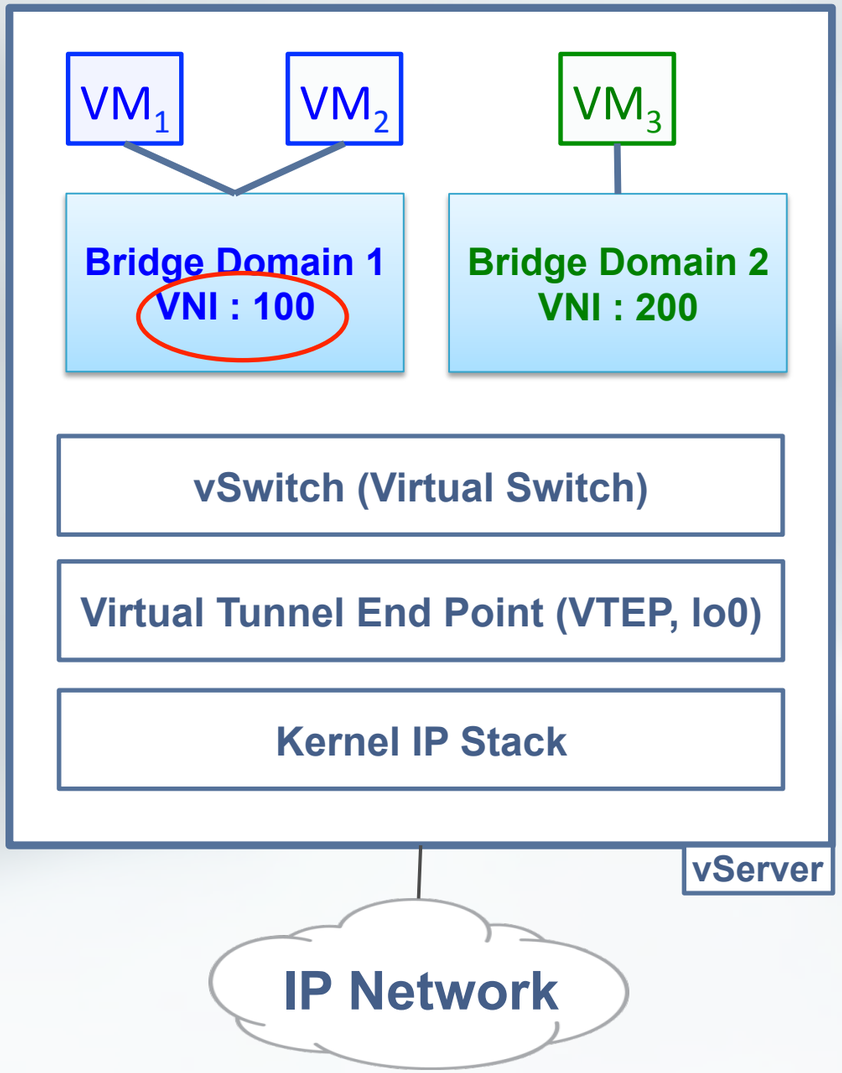


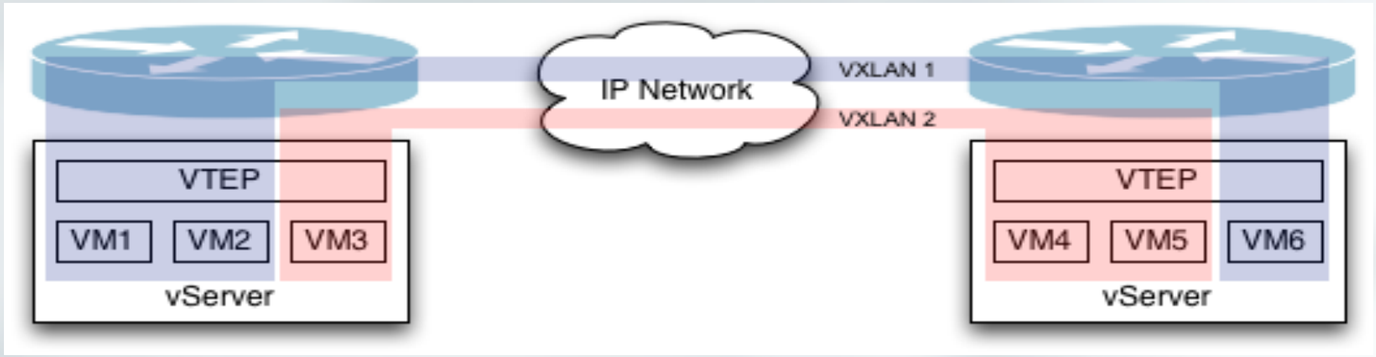
Overlay, Underlay, Gateway

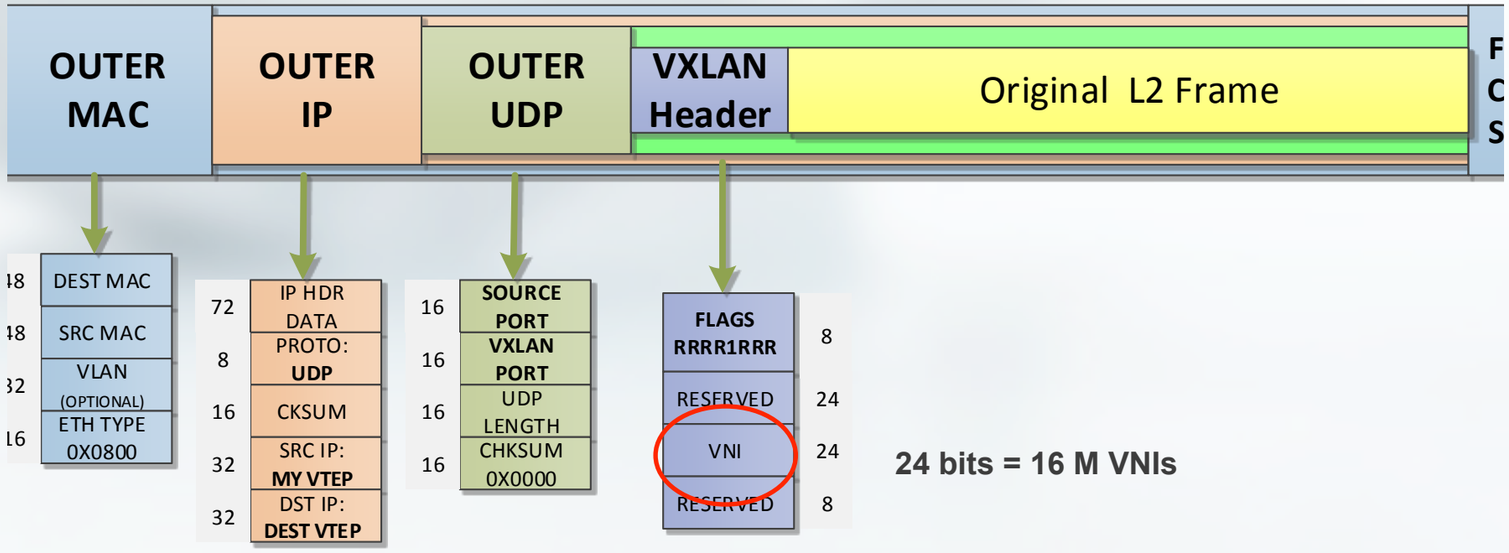
- VTEP間でVXLANトンネルを接続
- 同一VLANはフルメッシュ
- コントロールプレーンは、、、

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VXLAN : Building blocks



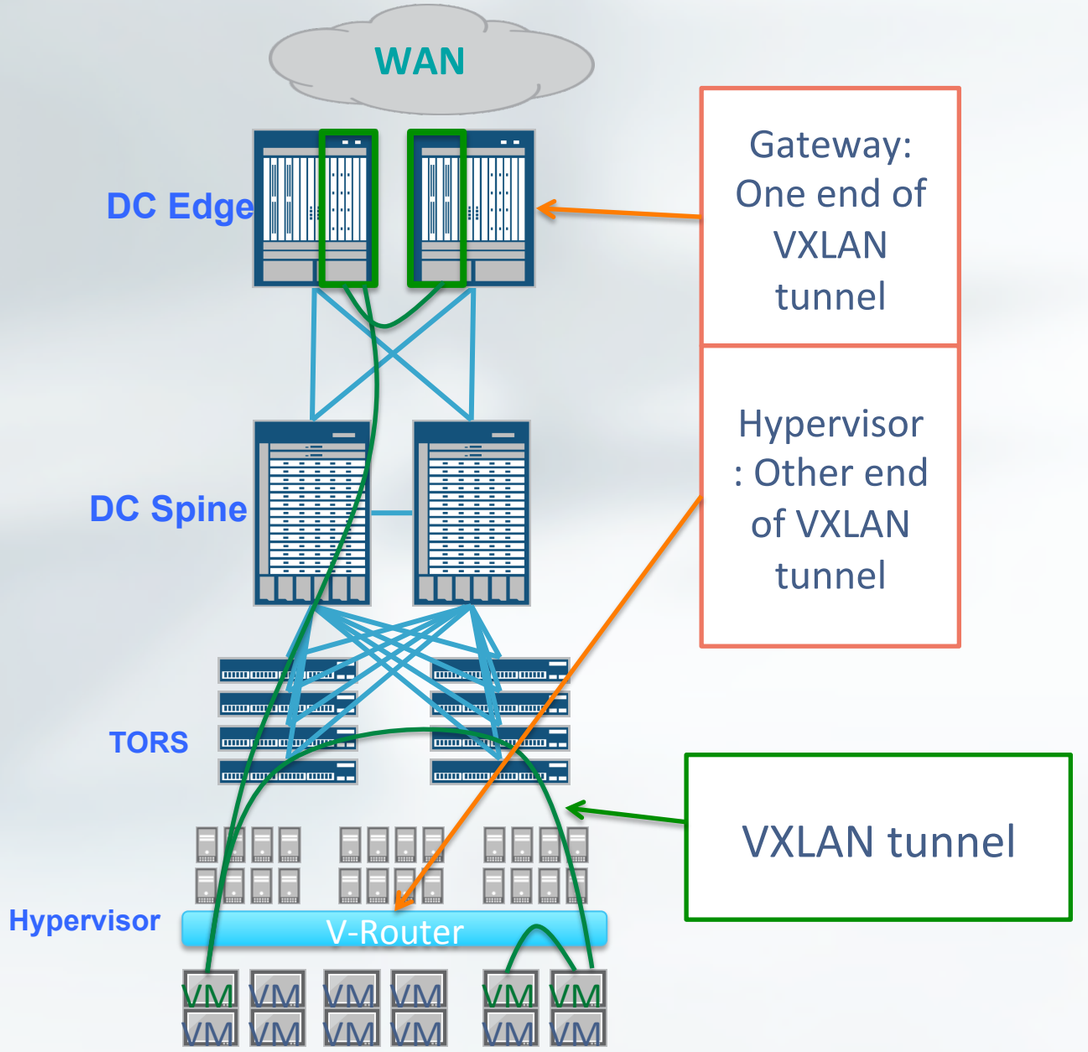




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VXLAN実装のオプション – IPマルチキャスト vs EVPN

IPマルチキャスト	EVPN
仮想ネットワークをマルチキャスト(PIM)グループを作る	仮想ネットワークをEVPNで作る
• Known UnicastはFull mesh Tunnelを使う • L2 BUMはPIM Multicast Tunnelを使う	• EVPNでVNI, MACをDistribution • データプレーンはVXLAN Tunnel



EVPN/VXLANの利点
- UnderlayはIPルーティングに対応していれば良い
- VM mobilityやARP Proxyに対応している

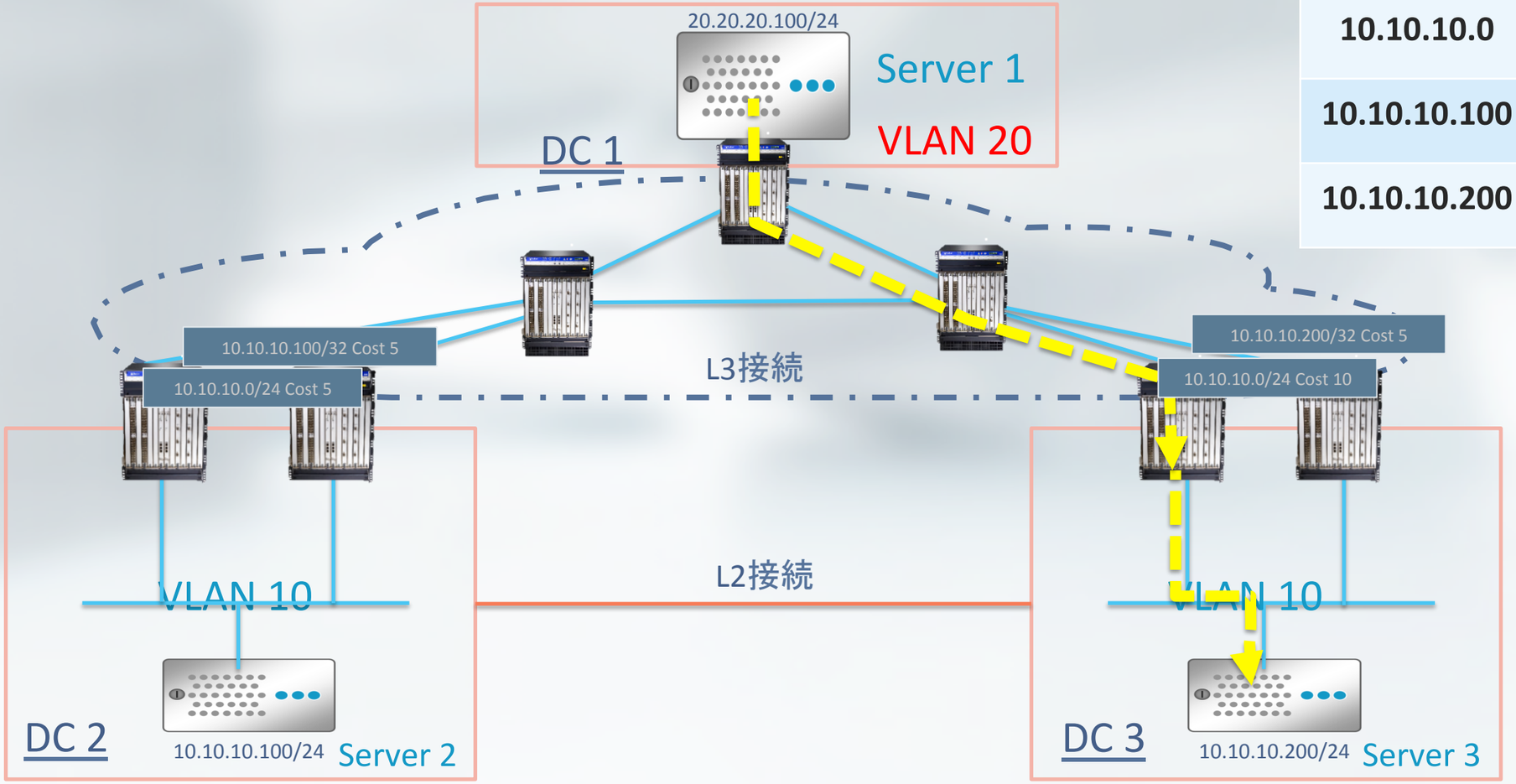
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EVPN VM MOBILITY: INGRESS TRAFFICの最適化

極端な例ですが、、
DC2 – DC3がL2で接続されて同一セグメントの場合、
DC1のServer 1は、DC3のServer 3にパケットを直接投げたい

DC 1's Edge Router Table

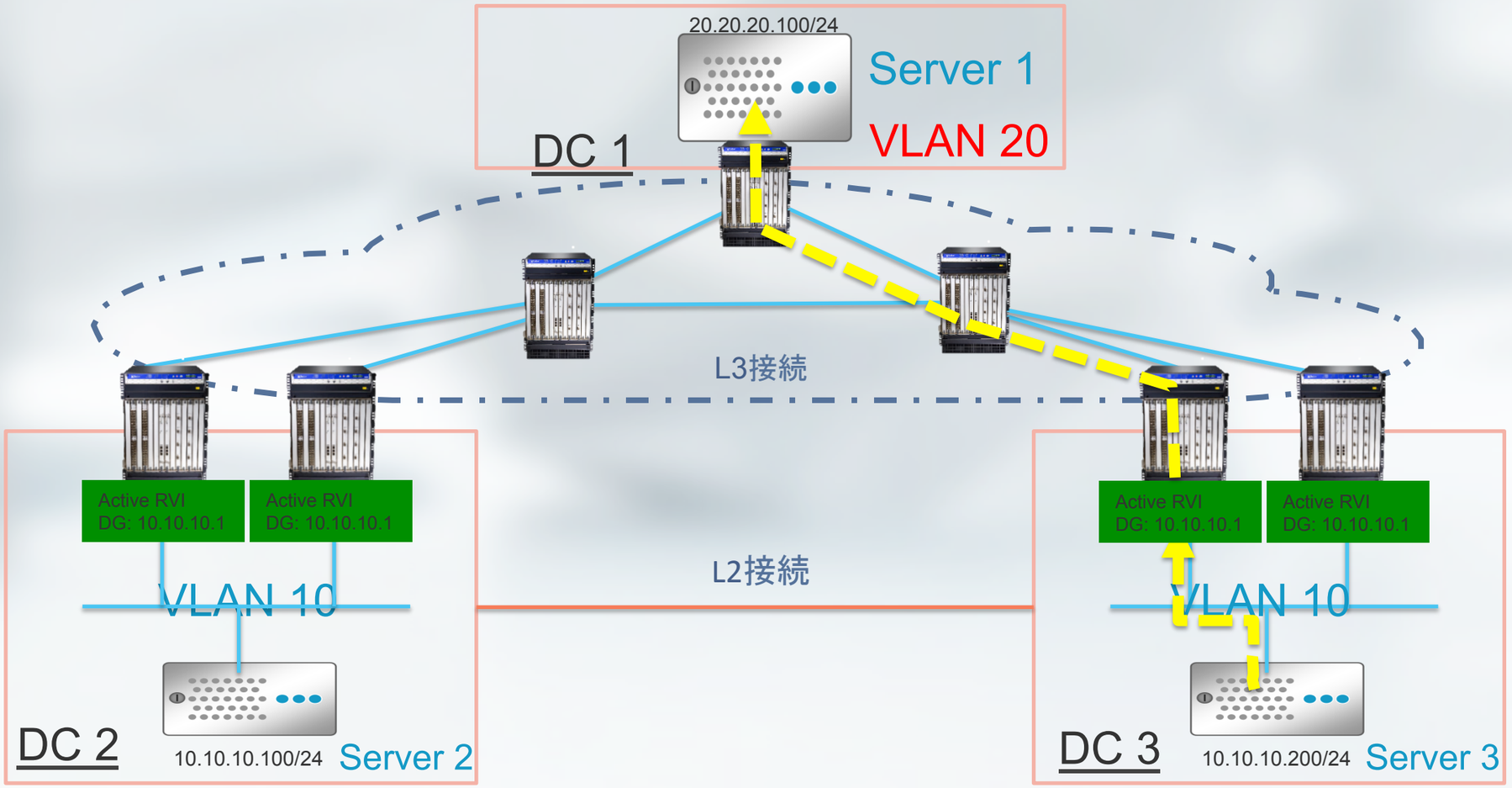
Route	Mask	Cost	Next Hop
10.10.10.0	24	5	Datacenter 2
10.10.10.0	24	10	Datacenter 3
10.10.10.100	32	5	Datacenter 2
10.10.10.200	32	5	Datacenter 3



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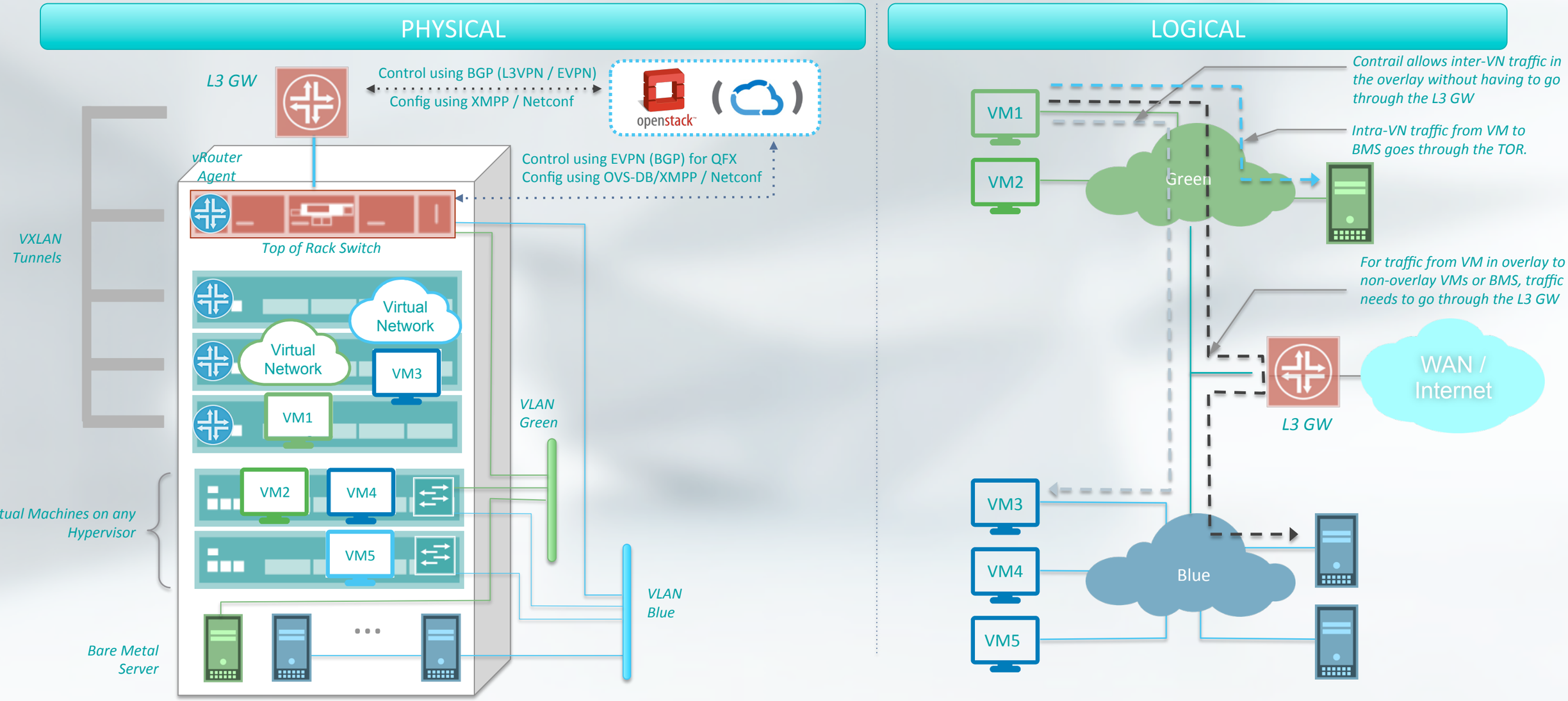
EVPN VM MOBILITY: EGRESS TRAFFICの最適化

Active-Activeで冗長化されたGWにパケットを送信



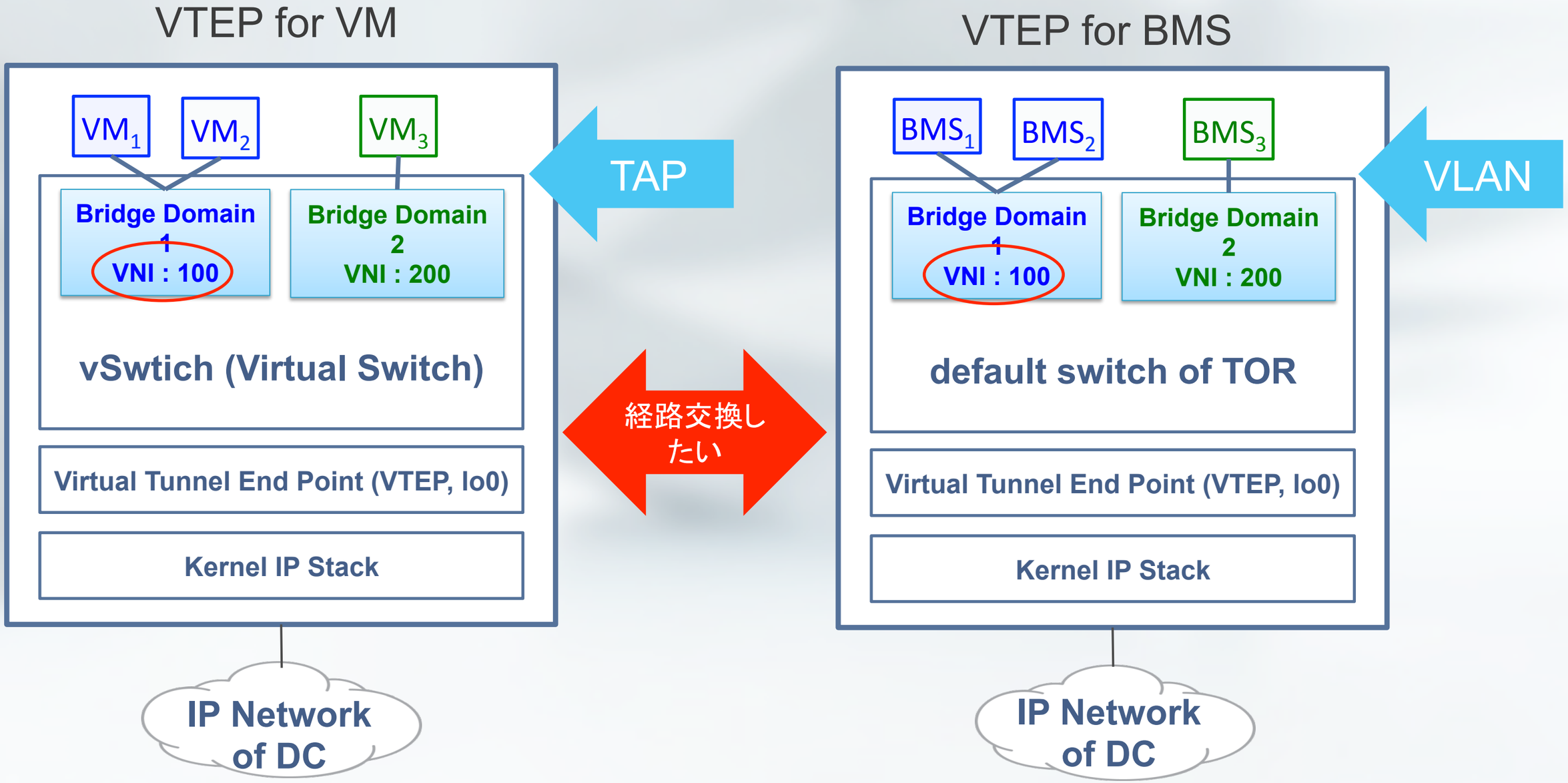
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VM and BMS INTEGRATION with TOR AND L3 GATEWAY



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VTEP for VM vs VTEP for Bare Metal Server



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VTEP for BMS – EVPN/VXLANのサンプルコンフィグ

```
# vlan configuration
v100 {
    vlan-id 100;
    vxlan {
        vni 100;
        ingress-node-replication;
    }
}
v101 {
    vlan-id 101;
    vxlan {
        vni 101;
        ingress-node-replication;
    }
}
# interface configuration
xe-0/0/0
unit 0 {
    family ethernet-switching {
        interface-mode trunk;
        vlan {
            members [ v100 v101 ];
        }
    }
}
```

```
# top level switch-options configuration
vtep-source-interface lo0.0;
route-distinguisher 1.2.3.11:1;
vrf-import imp;
vrf-target {
    target:1111:11;
    auto;
}
# top level protocol evpn configuration
encapsulation vxlan;
extended-vni-list [ 100 101 ]; ←or all

#policy configuration

policy-options policy-statement imp
term 1{
  from community [comglobal ];
  then accept;
}
term 100 {
  then reject;
}

community comglobal
members target:1111:11;
```


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VTEP for BMS – EVPN/VXLANデータプレーン　CLI出力例

```
root@node11# run show vlans

Routing instance        VLAN name               Tag         Interfaces
default-switch          default                 1
                                                            ae0.0*
                                                            esi.1746*
                                                            vtep.32769*
                                                            vtep.32770*
                                                            xe-0/0/0.0*
default-switch          v100                    100
                                                            ae0.0*
                                                            esi.1746*
                                                            vtep.32769*
                                                            vtep.32770*
default-switch          v101                    101
                                                            ae0.0*
                                                            esi.1746*
                                                            vtep.32769*
                                                            vtep.32770*
default-switch          v102                    102
```


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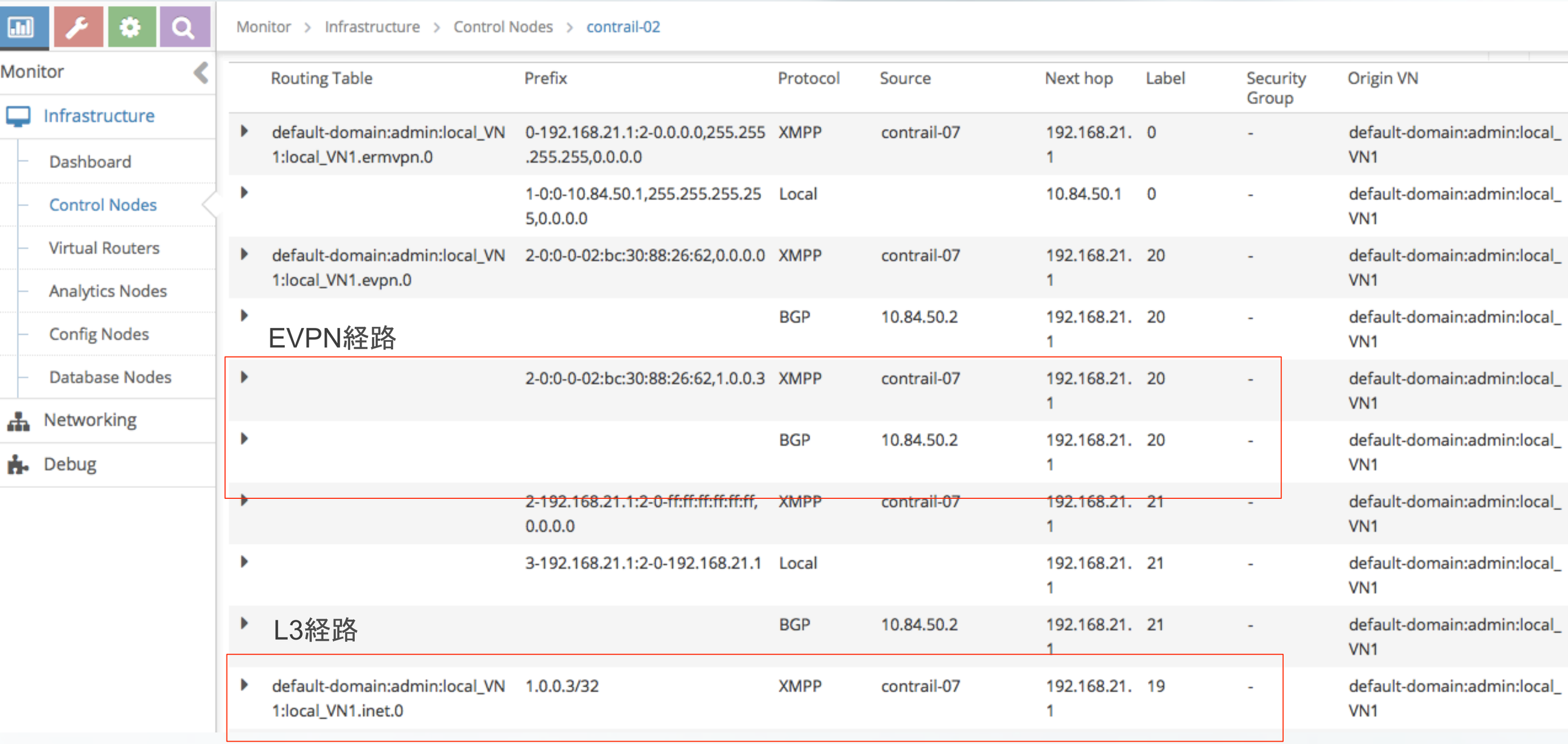
VTEP for BMS – EVPN/VXLANコントロールプレーン　CLI出力例

```
root@node11# run show route table default-switch.evpn.0

1:1.2.3.11:1::0101010101010101::0/304
                   *[EVPN/170] 04:12:55
                      Indirect
1:1.2.3.12:0::0101010101010101::FFFF:FFFF/304
                   *[BGP/170] 04:12:41, localpref 100, from 1.2.3.12
                      AS path: I, validation-state: unverified
                    > to 112.1.1.1 via xe-0/0/1.0
2:1.2.3.12:1::100::00:00:0a:0b:0c:0d/304
                   *[BGP/170] 00:00:06, localpref 100, from 1.2.3.12
                      AS path: I, validation-state: unverified
                    > to 112.1.1.1 via xe-0/0/1.0
3:1.2.3.11:1::100::1.2.3.11/304
                   *[EVPN/170] 05:27:51
                      Indirect
3:1.2.3.12:1::100::1.2.3.12/304
                   *[BGP/170] 04:13:31, localpref 100, from 1.2.3.12
                      AS path: I, validation-state: unverified
                    > to 112.1.1.1 via xe-0/0/1.0
```


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VTEP for VM – vRouterのRouting Table L3VPNとEVPNの活用

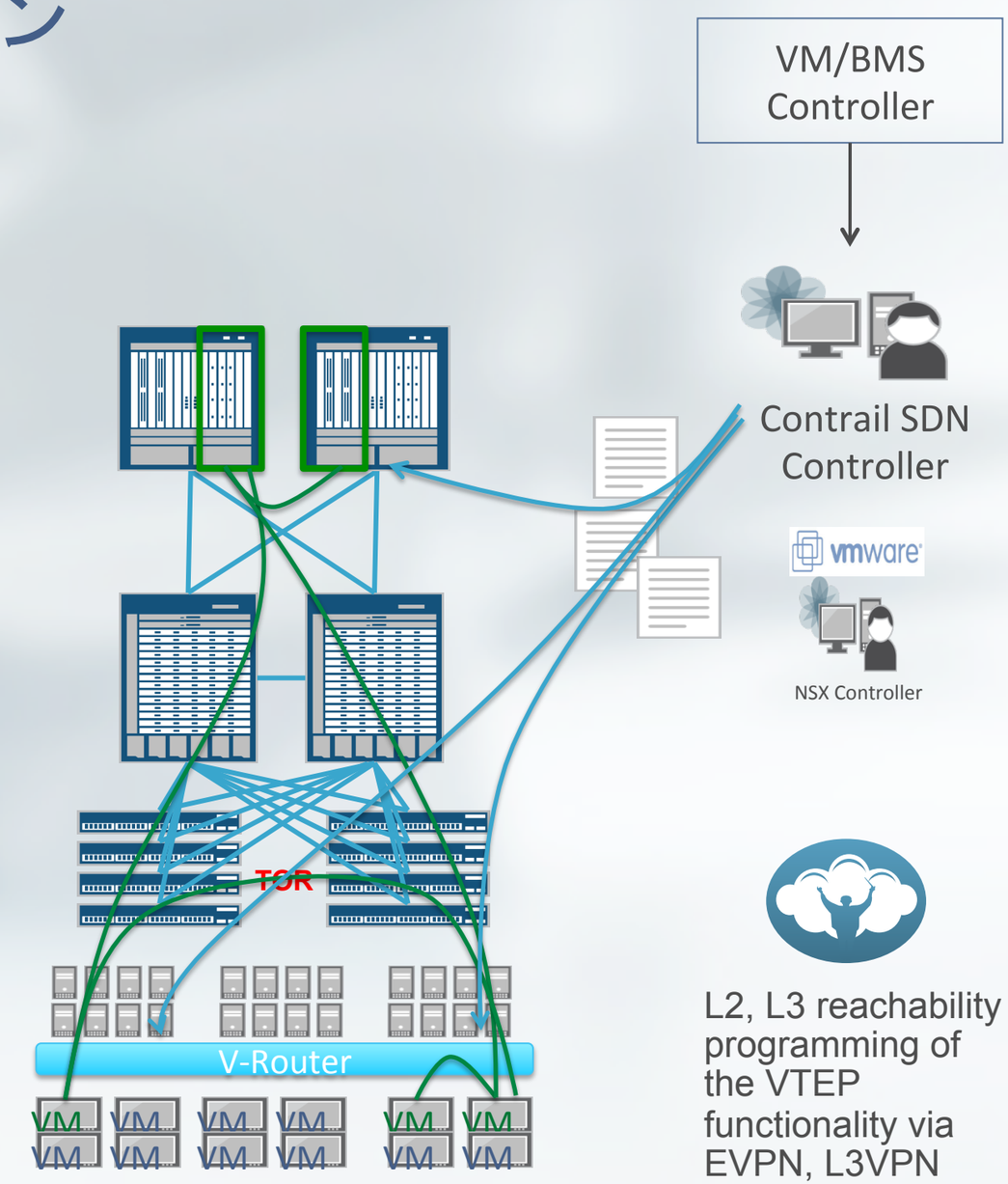


Routing Table	Prefix	Protocol	Source	Next hop	Label	Security Group	Origin VN
default-domain:admin:local_VN1:local_VN1.ermvpn.0	0-192.168.21.1:2-0.0.0.0,255.255.255.255,0.0.0.0	XMPP	contrail-07	192.168.21.1	0	-	default-domain:admin:local_VN1
	1-0:0-10.84.50.1,255.255.255.255,0.0.0.0	Local		10.84.50.1	0	-	default-domain:admin:local_VN1
default-domain:admin:local_VN1:local_VN1.evpn.0	2-0:0-0-02:bc:30:88:26:62,0.0.0.0	XMPP	contrail-07	192.168.21.1	20	-	default-domain:admin:local_VN1
		BGP	10.84.50.2	192.168.21.1	20	-	default-domain:admin:local_VN1
	2-0:0-0-02:bc:30:88:26:62,1.0.0.3	XMPP	contrail-07	192.168.21.1	20	-	default-domain:admin:local_VN1
		BGP	10.84.50.2	192.168.21.1	20	-	default-domain:admin:local_VN1
	2-192.168.21.1:2-0-ff:ff:ff:ff:ff:ff,0.0.0.0	XMPP	contrail-07	192.168.21.1	21	-	default-domain:admin:local_VN1
	3-192.168.21.1:2-0-192.168.21.1	Local		192.168.21.1	21	-	default-domain:admin:local_VN1
		BGP	10.84.50.2	192.168.21.1	21	-	default-domain:admin:local_VN1
default-domain:admin:local_VN1:local_VN1.inet.0	1.0.0.3/32	XMPP	contrail-07	192.168.21.1	19	-	default-domain:admin:local_VN1

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EVPNを使ったSDNインテグレーション マルチベンダーTOR

- Contrail SDN controller
  - EVPNでMAC情報を交換
  - TORコンフィグ設定
    - TOR – OVSDB
      - OVS、QFXと接続実績
  - VMとBMSのProvisioning
    - North boundはREST API



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REST API Example – Logical Interface for BMS

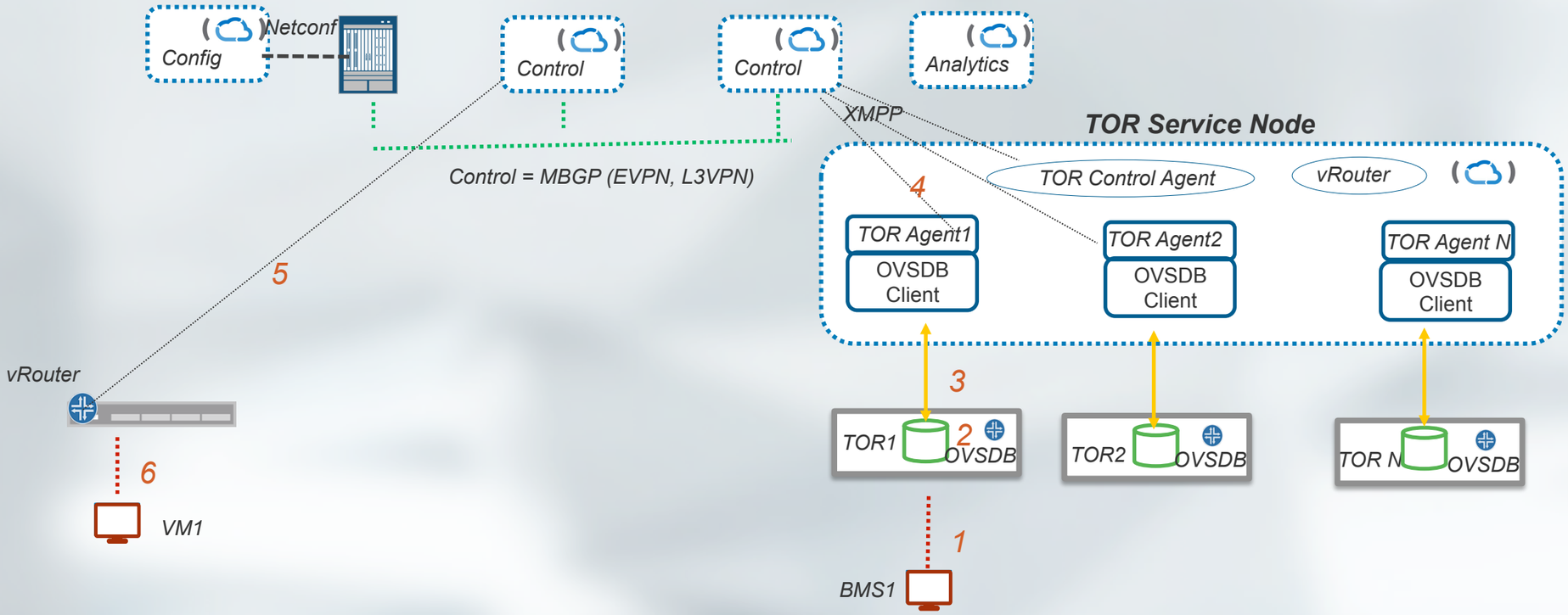
```
{
 u'logical-interface': {
   u'fq_name': [
    u'default-global-system-config’,
    u'test-router’,
    u'ge-0/0/1’,
    u'ge-0/0/1.0’
   ],
   u'parent_uuid': u'6608b8ef-9704-489d-8cbc-fed4fb5677ca’,
   u'logical_interface_vlan_tag': 0,
   u'parent_type': u'physical-interface’,
   u'virtual_machine_interface_refs': [
    {
u'to': [
          u'default-domain’,
          u'demo’,
          u'4a2edbb8-b69e-48ce-96e3-7226c57e5241'

]
```

Reference
http://www.juniper.net/techpubs/en_US/contrail2.1/topics/concept/rest-apis-routers-contrail.html

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MULTIVENDOR TOR solution – MAC LEARNING



1) RED BMS1 起動
2) OVSDBがBMS1用の新しいLocal MACをTOR1にアップデート
3) TOR1がTOR Service NodeのTOR Agent1に、BMS1 MACのNotification送信
4) TOR Agent1がControl NodeにMACを広告
5) Control Nodeが BMS1 MACを学習
6) 同様にVM1のMACがControl Node経由でTOR Agent1に広告され、TOR1にプログラムされる

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まとめ

EVPNのデータセンタでの活用

データセンタ技術との親和性

- 元々、VM mobilityやARP Proxyなどが考慮されている
- VXLAN VNI distributionをサポートしたので、DC向けスイッチに使いやすくなった

SDN技術との親和性

- SDNコントローラーのコントロールプレーンとして標準化された技術が使える
  - VXLANトンネルの接続先やMAC Learning
- データセンターで必要なVMやBMSのプロビジョニングと連携するユースケースで使いやすい

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everywhere